

# Title: PolyAlphabetic Cryptograph C codes

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## Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

// Encrypt function
void caesar_encrypt(char *plaintext, int shift, char *encrypted) {
    int i;
    for (i = 0; plaintext[i] != '\0'; i++) {
        char ch = plaintext[i];
        if (isalpha(ch)) {
            char base = isupper(ch) ? 'A' : 'a';
            encrypted[i] = (ch - base + shift) % 26 + base;
        } else {
            encrypted[i] = ch;
        }
    }
    encrypted[i] = '\0'; // Null terminator
}

// Decrypt function (reverse Caesar)
void caesar_decrypt(char *ciphertext, int shift, char *decrypted) {
    caesar_encrypt(ciphertext, 26 - shift, decrypted); // Reverse shift
}

// Simple hash: sum of ASCII values
int simple_hash(char *text) {
    int hash = 0;
    for (int i = 0; text[i] != '\0'; i++) {
        hash += text[i];
    }
    return hash;
}

int main() {
```

```

int choice;
char message[1000];
char output[1000];
int shift;

printf("Choose an option:\n");
printf(" [1] Encrypt\n");
printf(" [2] Decrypt\n");
printf("Enter choice: ");
scanf("%d", &choice);

printf("Enter your message: ");
getchar(); // Clear newline from buffer
fgets(message, sizeof(message), stdin);
message[strcspn(message, "\n")] = '\0'; // Remove newline

printf("Enter shift amount (1-25): ");
scanf("%d", &shift);

if (shift < 1 || shift > 25) {
    printf("Invalid shift. Must be between 1 and 25.\n");
    return 1;
}

if (choice == 1) {
    caesar_encrypt(message, shift, output);
    printf("\nEncrypted Message: %s\n", output);
    printf("Hash of Original Message: %d\n", simple_hash(message));
    printf("Hash of Encrypted Message: %d\n", simple_hash(output));
} else if (choice == 2) {
    caesar_decrypt(message, shift, output);
    printf("\nDecrypted Message: %s\n", output);
    printf("Hash of Encrypted Message: %d\n", simple_hash(message));
    printf("Hash of Decrypted Message: %d\n", simple_hash(output));
} else {
    printf("Invalid choice. Please enter 1 or 2.\n");
}

return 0;
}

```

## Sample Product:

| Output                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> Choose an option:  [1] Encrypt  [2] Decrypt Enter choice: 1 Enter your message: godwin Enter shift amount (1-25): 2  Encrypted Message: iqfypk Hash of Original Message: 648 Hash of Encrypted Message: 660  === Code Execution Successful === </pre> |

| Output                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> Choose an option:  [1] Encrypt  [2] Decrypt Enter choice: 2 Enter your message: iqfypk Enter shift amount (1-25): 2  Decrypted Message: godwin Hash of Encrypted Message: 660 Hash of Decrypted Message: 648  === Code Execution Successful === </pre> |

